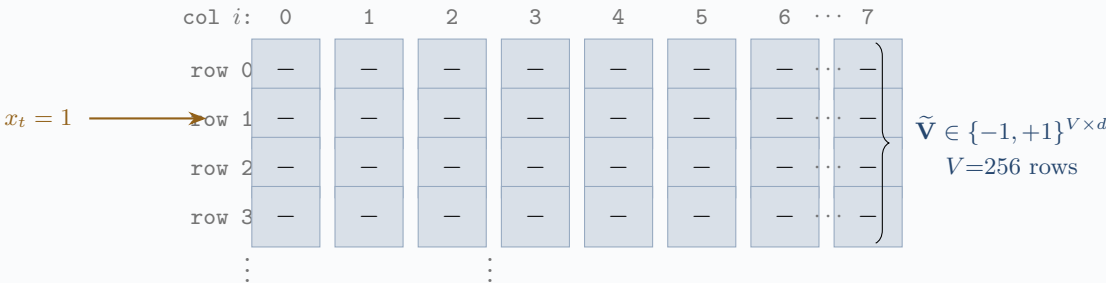


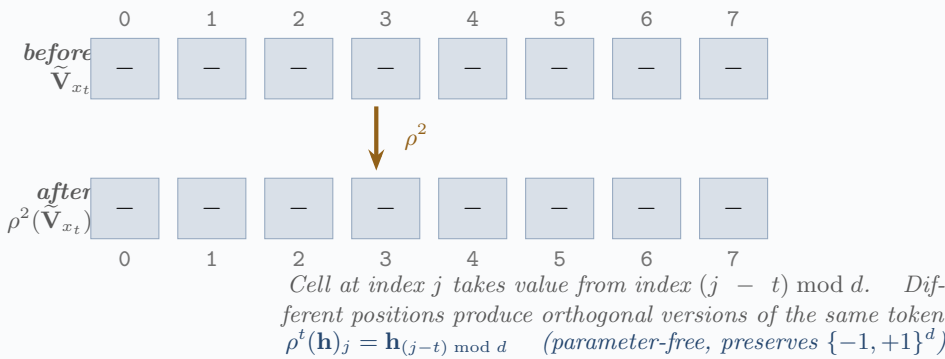
HDC-RWKV (nb12c) — Full Architecture Expansion

$V=256, d=256, L=1$, block size 16 • 16,434 deployment bytes • every learnable parameter bipolar at inference

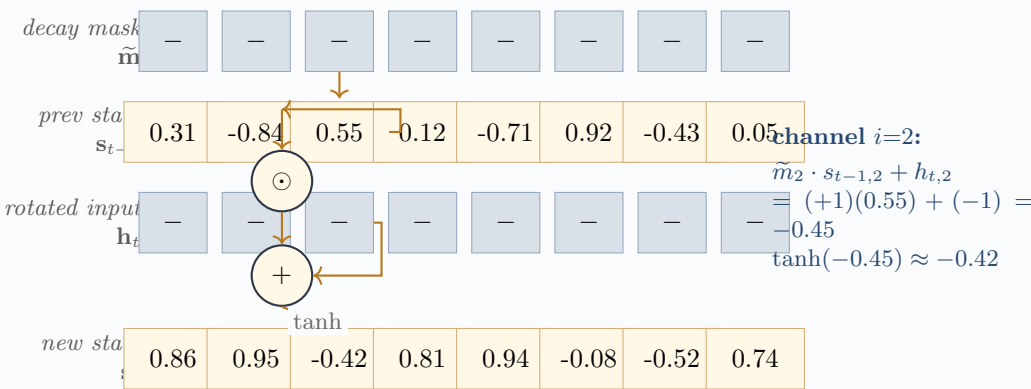
Block 1. Token lookup — bipolar embedding table



Block 2. Position binding ρ^t (cyclic right-rotation)

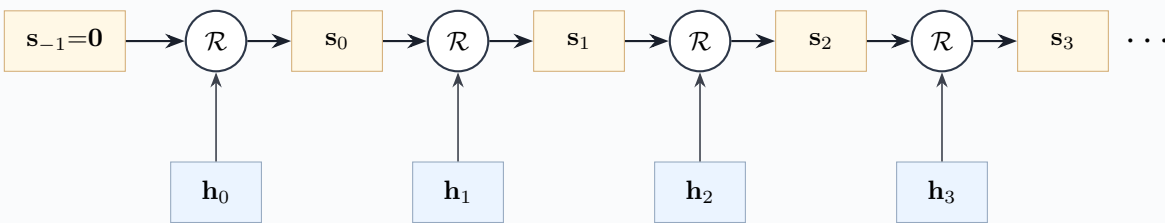


Block 3a. Recurrence — per-channel detail at one timestep t



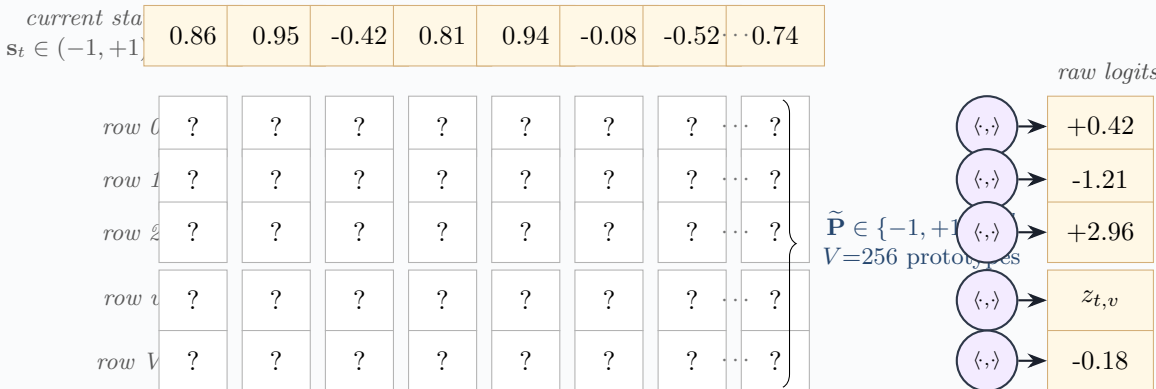
$$\mathbf{s}_t = \tanh(\tilde{\mathbf{m}} \odot \mathbf{s}_{t-1} + \mathbf{h}_t) \quad (\text{one channel at a time})$$

Block 3b. Recurrence — unrolled across time

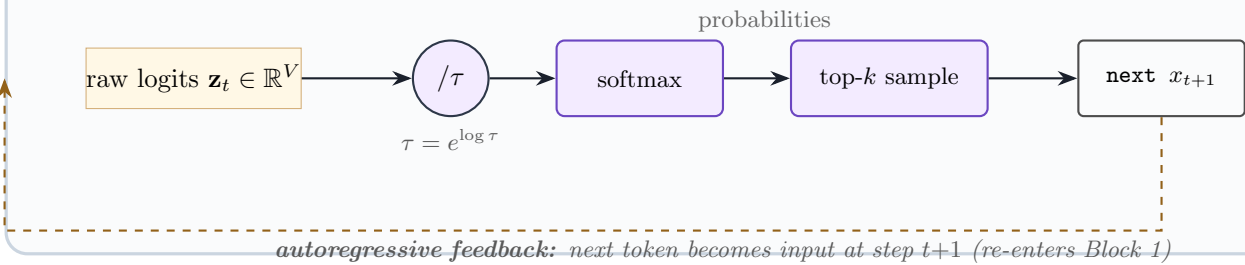


$\mathcal{R}(\mathbf{s}_{t-1}, \mathbf{h}_t) = \tanh(\tilde{\mathbf{m}} \odot \mathbf{s}_{t-1} + \mathbf{h}_t) \quad O(d) \text{ work per step. No attention. No key-value cache.}$

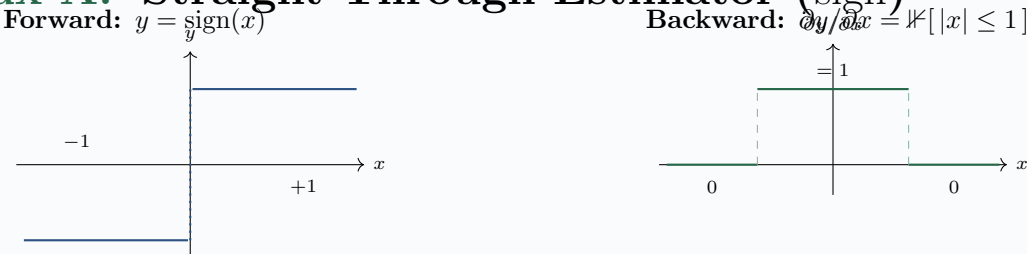
Block 4a. Prototype-similarity output — one dot product per vocabulary item



Block 4b. Temperature scaling, softmax, top- k sampling, autoregressive feedback



Aux A. Straight-Through Estimator (sign)



Forward is hard binarisation; backward is identity on the active strip $[-1, 1]$, zero outside. The optimiser updates the continuous shadow $\tilde{\mathbf{V}}$ while the recurrence always sees $\tilde{\mathbf{V}} \in \{-1, +1\}^{V \times d}$.

Aux B. Deployment equivalent — XOR + popcount

Encode $-1 \mapsto 0, +1 \mapsto 1$. Then for $a, b \in \{-1, +1\}^d$:

$$\langle a, b \rangle = d - 2 \cdot \text{popcount}(\bar{a} \oplus \bar{b}).$$

